Week 7 Cost-Benefit Memo

To	Dr. De Freitas, Mercer General ED Board, and Martina Griffith, Clinical IT
From	Terry Benjamin Jr.
Date	20 July 2026
Subject	Model choice for the Phase 3 emergency-triage prototype

I recommend carrying the tuned logistic model into Phase 3 as a clinician-facing shadow-mode comparator, not as an autonomous triage tool.

Dataset and method

I used the 55,121-visit Yale emergency-triage dataset and kept the Week 6 split unchanged: 44,096 training visits and 11,025 test visits, including 16 ESI 1 visits. The logistic row uses 209 numeric features. The complex models add 11 transparent features across circulation, breathing, temperature, and glucose.

Random Forest and histogram gradient boosting were tuned with three-fold cross-validation on the training set. The holdout was not used to choose settings. Macro precision, recall, and F1 average the five ESI classes equally. Inference time is the median of seven full-holdout runs divided by 11,025 patients.

Seven-dimension benchmark

Model	Acc	Prec	Recall	F1	ESI 1	Train s	Infer ms	Explain
Week 6 tuned logistic	0.681	0.553	0.511	0.501	7/16	3.36	0.0012	High
Tuned Random Forest	0.572	0.425	0.536	0.444	5/16	4.98	0.0078	Medium
Tuned gradient boosting	0.623	0.460	0.588	0.485	6/16	28.63	0.0153	Low

The strongest complex candidate was Tuned gradient boosting. Against the tuned logistic model, it changed macro F1 by -0.0162 and ESI 1 recall by -6.2%. It took 8.5 times as long to train and 13.3 times as long to infer, although every inference result remained below one millisecond per patient on this machine.

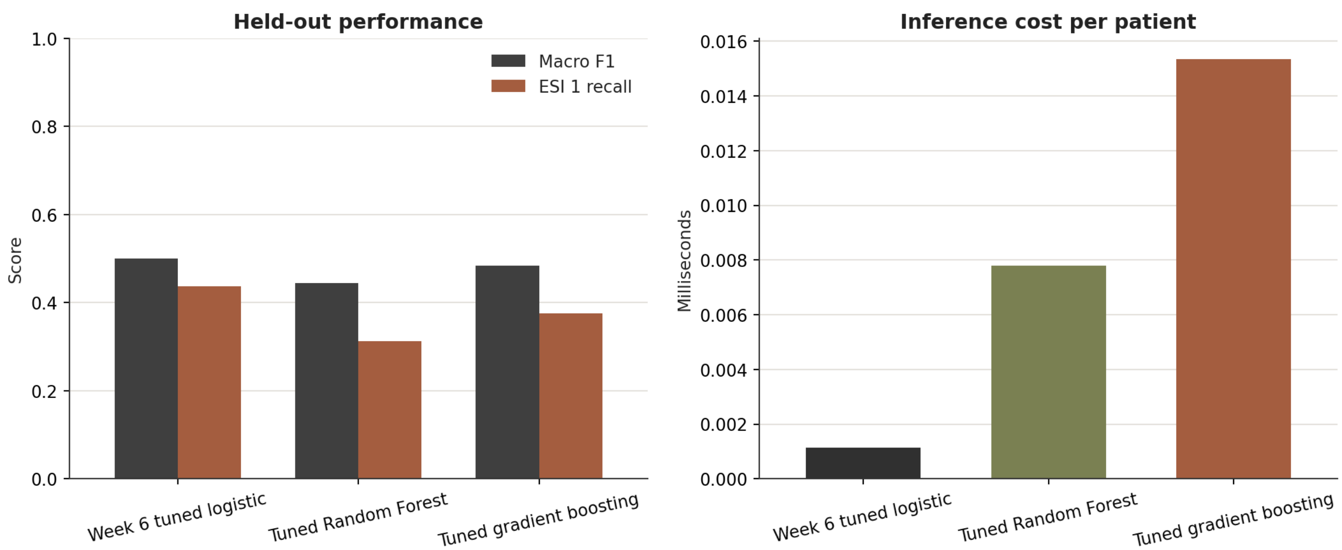

Figure 1. Held-out macro F1 and ESI 1 recall alongside median inference cost per patient. Timing is a relative laptop benchmark, not a production capacity test.

Why this is the most defensible choice

Three arguments for

1. Urgent-class signal. It caught 7 of 16 ESI 1 visits (43.8%), compared with 6 for the strongest complex candidate.

2. Marginal benefit. The strongest complex model changed macro F1 by only -0.016 and ESI 1 recall by -6.2%. That is not a material clinical gain.

3. Explanation and governance. Its scaled coefficients give a direct, reviewable account of which features pushed a prediction towards each ESI class. The simpler dependency chain also reduces the number of components that Clinical IT must version, monitor, and audit.

Three arguments against

1. ESI 1 performance is inadequate. It still missed 9 of 16 ESI 1 visits. That failure rate prevents any autonomous use.

2. The model form is restrictive. A linear decision boundary can miss interactions among complaints, oxygen status, age, and vital-sign combinations that tree ensembles can represent.

3. Rare-class evidence is thin. The recommendation rests on only 16 ESI 1 test visits. One case moves recall by 6.25 percentage points, so the ranking is not stable enough for a deployment decision.

What the comparison does and does not prove

The benchmark supports a relative model choice on one fixed retrospective holdout. It does not establish clinical effectiveness. A small timing difference on one laptop cannot stand in for a production capacity test, and a macro score cannot stand in for the consequences of one under-triaged patient.

The Random Forest's global feature importance is useful for checking what the ensemble relies on. It is not a faithful explanation of one patient's prediction. Gradient boosting has the same problem more strongly and would need an additional local explanation method before clinician review.

Cost and benefit in practice

The choice is not being driven by raw speed. All three models predicted the 11,025-patient holdout in well under one millisecond per patient on the test machine. The meaningful cost is the work needed to understand, validate, monitor, and defend each model. Random Forest and gradient boosting can represent nonlinear interactions, but neither converted that flexibility into a better combination of macro F1 and ESI 1 recall.

The logistic model's benefit is practical rather than absolute. It gives the strongest current urgent-class result, a direct coefficient-based review path, and a stable reference for later models. Its cost is equally clear: it cannot express complex interactions and its nine missed ESI 1 visits make it unsuitable for autonomous use.

For Phase 3, a replacement should improve both ESI 1 recall and macro F1 on external or prospective data, not merely accuracy or one common class. It must also provide patient-level explanations that clinicians can review and remain within the same workflow and latency constraints.

Clinical error review and Phase 3 recommendation

Critical ESI 1 errors

The recommended model caught 7 of the 16 held-out ESI 1 visits and under-triaged 9. Under the notebook's transparent near-normal screen, 3 of 9 missed visits had heart rate 60-100, respiratory rate 12-20, oxygen saturation at least 94%, systolic pressure at least 100, and temperature 96.8-100.4 F. The same was true for 3 of 7 caught visits.

The most common complaint flags among misses were other (4 missed), shortness of breath (2 missed), altered mental status (1 missed), dizziness (1 missed), hypotension (1 missed). These aggregate counts do not show causation. They identify a specific feature and data question: critical presentations with ordinary-looking arrival vitals may need more complaint context than a broad binary flag provides.

Risks and unknowns

1. External validity. The data come from a Yale emergency department and have not been tested at Mercer General or another Caribbean ED.

2. Rare-class uncertainty. Only 16 ESI 1 visits are in the holdout, so class-specific recall is unstable.

3. Feature and label limits. Recorded ESI can vary by local practice, and the table does not capture all free-text nuance, clinician concern, medication, or oxygen context.

4. Unfinished safety work. Calibration, subgroup performance, drift, prospective latency, and human-factors testing remain undone.

Phase 3 evidence plan

1. Local validation. Freeze the feature definitions and test on a representative Mercer General sample. Report raw ESI 1 counts and uncertainty, not only percentages.

2. Safety review. Review under-triaged cases with ED clinicians, test calibration and subgroup performance, and investigate oxygen support, medication, and complaint detail as explanations for recurring misses.

3. Shadow workflow test. Run beside nurse-led triage and measure data failures, latency, disagreement, and override handling without putting model output into the live ESI decision.

Recommendation

Carry Week 6 tuned logistic into Phase 3 only as a shadow-mode decision support comparator. It should run behind nurse-led triage and must not delay, replace, or automatically downgrade a clinician's ESI decision.

Before a live pilot, collect more ESI 1 examples, validate on local data, test calibration and subgroup performance, review the missed complaint patterns with ED clinicians, and define fail-safe handling for missing or out-of-range inputs. Replace this model only if a complex candidate shows a repeatable, clinically material ESI 1 and macro-F1 gain on external or prospective data while meeting the same explanation and latency requirements.

Evidence

Notebook: notebooks/week7_final_model_tradeoff.ipynb | Benchmark: docs/week7_final_benchmark.md | Decision: docs/decisions/2026-week-7-model-choice.md